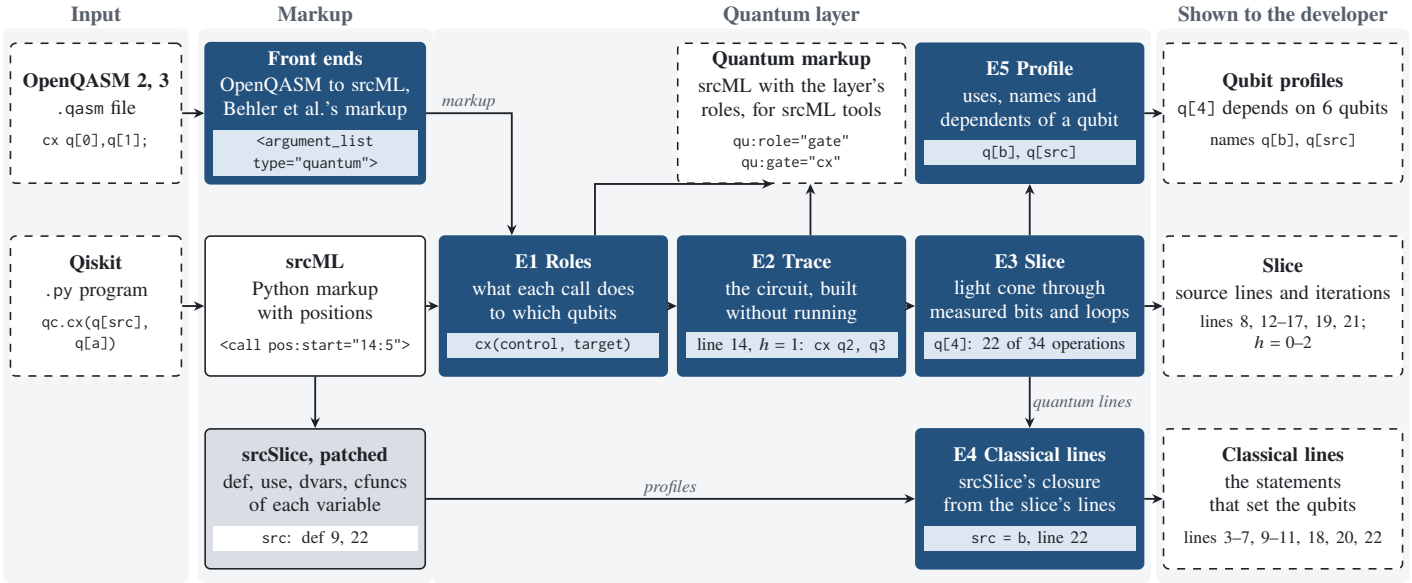

Input
Markup
Quantum layer
Shown to the developer
OpenQASM 2, 3
.qasm file
cx q[0],q[1];
Front ends
OpenQASM to srcML, Behler et al.'s markup
<argument_list type="quantum">
markup
Quantum markup
srcML with the layer's roles, for srcML tools
qu:role="gate"
qu:gate="cx"
E5 Profile
uses, names and dependents of a qubit
q[b], q[src]
Qubit profiles
q[4] depends on 6 qubits
names q[b], q[src]
Qiskit
.py program
qc.cx(q[src], q[a])
srcML
Python markup with positions
<call pos:start="14:5">
E1 Roles
what each call does to which qubits
cx(control, target)
E2 Trace
the circuit, built without running
line 14, h = 1: cx q2, q3
E3 Slice
light cone through measured bits and loops
q[4]: 22 of 34 operations
Slice
source lines and iterations
lines 8, 12–17, 19, 21; h = 0–2
quantum lines
srcSlice, patched
def, use, dvars, cfuncs of each variable
src: def 9, 22
profiles
E4 Classical lines
srcSlice's closure from the slice's lines
src = b, line 22
Classical lines
the statements that set the qubits
lines 3–7, 9–11, 18, 20, 22
taken unchanged
extended and fixed
new
input or output
each part shows its result on the program of Fig. 1, for q[4]